

AI / ML Training Program

60-Day Syllabus

Prepared for

Infiway Global

60	6	6	100%
Days	Domains	Portfolio Projects	Hands-On

1. Program at a Glance

Duration	60 days
Daily class time	1 to 2 hours of instructor-led training
Cohort	Mixed - technical and non-technical students
End goal	Entry-level AI/ML job-readiness
Approach	Project-driven; every concept ends in working, deployed code
Deliverables	6 portfolio projects, all deployed publicly
Final outcomes	GitHub profile, deployed apps, resume, LinkedIn, mock-interview clearance

2. What Students Will Learn

Six domains, taught in sequence so each builds on the previous one.

#	Domain	Core topics covered
1	Foundations	Python, NumPy, Pandas, Matplotlib, Seaborn, statistics, linear algebra basics, Git & GitHub
2	Classical Machine Learning	Regression, classification, clustering, decision trees, Random Forest, XGBoost, feature engineering, model evaluation, hyperparameter tuning
3	Deep Learning	Neural networks, Keras / TensorFlow, CNNs, RNNs, LSTMs, transfer learning, regularization
4	Computer Vision	OpenCV, image processing, YOLO object detection, image segmentation, MediaPipe pose & face tracking
5	NLP & GenAI / LLMs	Text processing, embeddings, transformers, Hugging Face, prompt engineering, LLM APIs, RAG (Retrieval-Augmented Generation)
6	MLOps & Deployment	FastAPI, Streamlit, Docker, cloud deployment, MLflow, basic CI/CD with GitHub Actions

3. The 60-Day Breakdown

Each day is broken into: topic taught, what's done in class, the homework/assignment, and the output the student will have by end of day.

Phase 1 - Foundations (Day 1 to 12)

Day 1 - Course Kickoff, AI/ML/DL Landscape, Tools Setup

- **Topic taught:** What AI, ML, DL, and Data Science actually are with real-world examples; the 5 career tracks (Data Analyst, ML Engineer, Data Scientist, AI Engineer, MLOps Engineer); the 60-day roadmap
- **In class:** Live installation of Anaconda, VS Code, Jupyter; create GitHub account, first repo, first commit; run Hello World in Jupyter
- **Homework:** Push a README.md to GitHub with your name, background, and why you want to learn AI/ML; submit the link
- **Output:** Working environment ready + first GitHub commit live

Day 2 - Python Essentials I - Variables, Types, Operators, I/O

- **Topic taught:** The 4 core data types (int, float, string, bool); reading user input; printing formatted output; arithmetic, comparison, and logical operators
- **In class:** 10 micro-problems solved live (calculator, age checker, BMI calculator, etc.)
- **Homework:** Solve 20 problems on HackerRank in the Python easy tier
- **Output:** HackerRank screenshots showing solved problems submitted

Day 3 - Python Essentials II - Lists, Tuples, Dictionaries, Sets

- **Topic taught:** When to use each data structure; list operations; dictionary operations; list comprehensions
- **In class:** Build a working student-records system using only lists and dicts (add, search, sort, average marks)
- **Homework:** Build a CLI Expense Tracker with add expense, view all, total by category, sort by amount; push to GitHub
- **Output:** Expense tracker repo live on GitHub

Day 4 - Python Essentials III - Loops, Functions, Modules

- **Topic taught:** for and while loops with break and continue; writing functions; default and keyword arguments; importing from standard library (math, random, datetime)
- **In class:** Refactor Day 3 expense tracker into clean reusable functions

- **Homework:** Build mystats.py - a custom module with functions for mean, median, mode, standard deviation (no imports allowed); write 10 test cases
- **Output:** Custom stats library on GitHub with passing tests

Day 5 - NumPy I - Arrays, Indexing, Broadcasting

- **Topic taught:** Why NumPy is faster than lists (speed demo); creating arrays (zeros, ones, arange, linspace); 1D and 2D indexing & slicing; broadcasting basics
- **In class:** Solve 8 NumPy problems together - normalize array, find max along axis, create masks
- **Homework:** Solve 15 more NumPy problems + create a checkerboard pattern as a NumPy array (visual challenge)
- **Output:** NumPy problems notebook + checkerboard image

Day 6 - NumPy II - Linear Algebra & Stats Essentials

- **Topic taught:** Vectors, matrices, dot product, transpose; descriptive statistics (mean, median, std, variance, percentiles) using NumPy
- **In class:** Compute summary statistics on a sales CSV using only NumPy (no Pandas yet)
- **Homework:** Write a NumPy-only mini-EDA on the Iris dataset; no Pandas, no Matplotlib, just NumPy + print
- **Output:** Iris EDA in pure NumPy on GitHub

Day 7 - Pandas I - Series, DataFrames, Loading Data

- **Topic taught:** What a DataFrame is; loading CSV/Excel/JSON; inspecting data with .head, .info, .describe, .shape, .dtypes; selecting rows and columns with loc and iloc
- **In class:** Load the Titanic dataset and answer 10 inspection questions (how many passengers, average age, gender split, etc.)
- **Homework:** Load 2 datasets of your choice from Kaggle; write a 1-page 'first impressions' notebook for each
- **Output:** 2 inspection notebooks on GitHub

Day 8 - Pandas II - Cleaning, Merging, Groupby

- **Topic taught:** Handling missing values (drop vs fill); removing duplicates; merging and joining DataFrames; groupby and pivot tables
- **In class:** Live cleaning of a deliberately dirty dataset - fixing missing values, wrong data types, duplicates
- **Homework:** Take Kaggle's Adult Income dataset, clean it, and write 5 insights using groupby (income by education, occupation, gender, etc.)
- **Output:** Cleaned dataset notebook with 5 documented insights

Day 9 - Visualization - Matplotlib & Seaborn

- **Topic taught:** Which chart for which question (line for trends, bar for categories, scatter for relationships, heatmap for correlations); building publication-quality plots
- **In class:** Build 5 charts on Titanic that answer 5 specific questions
- **Homework:** Build 10 charts on a dataset of your choice; write a one-line insight under each chart
- **Output:** 10-chart visualization notebook

Day 10 - Statistics & Probability Essentials

- **Topic taught:** Mean vs median vs mode and when each lies; normal vs skewed distributions; variance, standard deviation, z-scores; basic probability
- **In class:** Generate a normal distribution in NumPy, compute z-scores, identify outliers
- **Homework:** Build a Z-Score Outlier Detector - input any CSV column, return rows with z-score above 3
- **Output:** Outlier detector script on GitHub

Day 11 - Linear Algebra & Calculus Intuition

- **Topic taught:** Lines, slopes, the geometric meaning of regression; derivatives as 'rate of change' (no formulas); gradients explained as 'climbing a hill' - pure intuition, no proofs
- **In class:** Manually fit a line to 5 data points by trying different slopes; minimize squared error by hand
- **Homework:** Watch one 3Blue1Brown video on linear algebra; write 3 sentences on what clicked
- **Output:** Reflection note + completed line-fitting exercise

Day 12 - PROJECT 1 - Exploratory Data Analysis on a Real Dataset

- **Topic taught:** The complete EDA workflow - load, clean, visualize, summarize, narrate
- **In class:** Project briefing; students choose ONE dataset: Airbnb listings, IMDb movies, NYC Yellow Taxi, or Indian Census
- **Homework:** Complete the EDA notebook tonight - 8-10 charts + markdown narrative + clean README; push to GitHub
- **Output:** PORTFOLIO PROJECT 1 - complete EDA notebook live on GitHub

Phase 2 - Classical Machine Learning (Day 13 to 28)

Day 13 - ML Mental Model & Workflow

- **Topic taught:** Supervised vs unsupervised vs reinforcement learning; the 8-step ML workflow; why we split train and test data
- **In class:** Load the Iris dataset, do a train/test split, train a simple model, measure accuracy
- **Homework:** Repeat the same workflow on 2 other sklearn toy datasets (Wine, Breast Cancer)
- **Output:** 3 baseline ML notebooks completed

Day 14 - Linear Regression (From Scratch + sklearn)

- **Topic taught:** Cost function (MSE); gradient descent intuition; implementing linear regression in NumPy; then using sklearn's LinearRegression
- **In class:** Build linear regression from scratch in ~30 lines of NumPy; compare with the sklearn version
- **Homework:** Apply linear regression to a housing price dataset; report MSE and R-squared score
- **Output:** Housing price model notebook

Day 15 - Ridge, Lasso, ElasticNet (Regularization)

- **Topic taught:** What overfitting is (with live demo); how regularization fixes it; difference between Ridge (L2), Lasso (L1), and ElasticNet
- **In class:** Show overfitting live, then add Ridge and watch validation error improve
- **Homework:** Tune alpha for Ridge and Lasso on the housing dataset; plot the regularization path
- **Output:** Regularization comparison notebook

Day 16 - Logistic Regression & Classification Metrics

- **Topic taught:** Logistic regression as 'regression squashed into a probability'; confusion matrix; accuracy, precision, recall, F1, ROC-AUC and when each one matters
- **In class:** Train logistic regression on the Pima Diabetes dataset; compute all metrics by hand from the confusion matrix
- **Homework:** Write a 1-page note on 'when is precision more important than recall' with 3 real industry examples (medical, spam, fraud)
- **Output:** Diabetes classifier + written metric analysis

Day 17 - Decision Trees

- **Topic taught:** How a tree splits (Gini, entropy); tree visualization; bias-variance tradeoff using tree depth
- **In class:** Train trees of depth 2, 5, 10, 20 on Titanic; plot train vs validation accuracy to see the overfitting curve

- **Homework:** Build a decision tree on a dataset of your choice; visualize the tree using `plot_tree`; embed the image in your README
- **Output:** Decision tree project with visualization on GitHub

Day 18 - Random Forest

- **Topic taught:** Bagging in one sentence; why many trees beat one tree; reading feature importances
- **In class:** Replace yesterday's decision tree with a Random Forest; compare results
- **Homework:** Submit Random Forest predictions to a beginner Kaggle competition; record your score
- **Output:** Kaggle submission with leaderboard screenshot

Day 19 - XGBoost & LightGBM

- **Topic taught:** Boosting vs bagging - how each tree fixes the previous tree's mistakes; why XGBoost wins most tabular Kaggle competitions
- **In class:** Train XGBoost with early stopping; beat the Random Forest baseline
- **Homework:** Tune XGBoost with at least 5 different hyperparameter combinations; document each result in a table
- **Output:** Hyperparameter tuning notebook with comparison table

Day 20 - KNN, Naive Bayes, SVM

- **Topic taught:** When to use each - KNN for small geometric problems, Naive Bayes for text, SVM for margin-based classification; kernel trick at intuition level only
- **In class:** Bake-off - train all 3 on the same small dataset and compare
- **Homework:** Compare KNN, Naive Bayes, and SVM on a dataset of your choice; write 100 words on which won and why
- **Output:** 3-model comparison study

Day 21 - Feature Engineering & Encoding

- **Topic taught:** One-hot, label, ordinal, target encoding - when to use which; engineering date features (day-of-week, month, is-weekend); binning numerical features
- **In class:** Live engineer 5 new features on a retail dataset and watch the model improve
- **Homework:** Re-engineer features on your Day 19 XGBoost project; document if it improved with before/after numbers
- **Output:** Updated XGBoost project with engineered features

Day 22 - Scaling, Imbalanced Data, Data Leakage

- **Topic taught:** StandardScaler vs MinMaxScaler; handling imbalanced data with class weights, SMOTE, undersampling; what data leakage is and how to spot it

- **In class:** Demo a deliberately leaky pipeline; students identify the bug
- **Homework:** Apply SMOTE to a fraud detection dataset; compare F1 score before and after
- **Output:** Fraud detection notebook with SMOTE results

Day 23 - Cross-Validation & Hyperparameter Tuning

- **Topic taught:** Why a single train/test split lies; k-fold cross-validation; GridSearchCV and RandomizedSearchCV; sklearn Pipelines that prevent leakage
- **In class:** Build a leak-free Pipeline with ColumnTransformer + GridSearchCV from scratch
- **Homework:** Build a Pipeline + GridSearch on your current best project; push to GitHub
- **Output:** Production-quality leak-free pipeline

Day 24 - K-Means & Hierarchical Clustering

- **Topic taught:** Customer segmentation use case; choosing K with elbow method and silhouette score; hierarchical clustering and dendrograms
- **In class:** Cluster a mall customers dataset; name each cluster as a persona (budget shoppers, luxury buyers, etc.)
- **Homework:** Build a Customer Segmentation Notebook with named personas and a 1-paragraph business recommendation
- **Output:** Customer segmentation notebook

Day 25 - PCA & Dimensionality Reduction

- **Topic taught:** PCA as 'shadow on a wall' - keeping the most important directions; explained variance; when NOT to use PCA; quick intro to t-SNE and UMAP
- **In class:** Apply PCA to the MNIST digits dataset; visualize in 2D
- **Homework:** Apply PCA to a high-dimensional dataset of your choice; visualize the top 2 components
- **Output:** PCA notebook with 2D visualization

Day 26 - PROJECT 2 - Day 1: Plan & EDA

- **Topic taught:** Project briefing; students choose ONE: Telco Churn, Loan Default, House Prices, or Employee Attrition
- **In class:** Define the business problem; plan the project; set up clean repo structure
- **Homework:** Complete EDA + first baseline model tonight
- **Output:** EDA notebook + first baseline model

Day 27 - PROJECT 2 - Day 2: Build & Tune

- **Topic taught:** Iteration - try 3+ models, tune the best one
- **In class:** Build session - train Random Forest, XGBoost, Logistic Regression; tune the winner

- **Homework:** Build a Streamlit app that takes user inputs and gives predictions; get it running locally
- **Output:** Working Streamlit app on local machine

Day 28 - PROJECT 2 - Day 3: Deploy & Present

- **Topic taught:** Deploying to Streamlit Community Cloud (free); writing a strong README
- **In class:** Live deployment walkthrough; each student presents project in 5 minutes (problem, data, model, results, demo link)
- **Homework:** Polish README; pin repo on GitHub profile
- **Output:** PORTFOLIO PROJECT 2 - live deployed ML web app

Phase 3 - Deep Learning (Day 29 to 40)

Day 29 - Neural Networks - Perceptron & Forward Propagation

- **Topic taught:** From logistic regression to neural networks one diagram at a time; perceptron, weights, bias, activation; manually computing the output of a 2-layer network
- **In class:** Forward propagation by hand on the whiteboard; then code it in NumPy
- **Homework:** Build a 2-layer neural network in NumPy that classifies the XOR problem; no Keras yet
- **Output:** XOR-solving NumPy NN on GitHub

Day 30 - Activations, Loss Functions, Gradient Descent

- **Topic taught:** Sigmoid, tanh, ReLU, softmax - when to use which; loss functions (MSE, binary cross-entropy, categorical cross-entropy); SGD vs Adam vs RMSprop
- **In class:** Experiment with different activations on a toy dataset; observe how training changes
- **Homework:** Build a binary classifier in raw NumPy on the 'moons' dataset
- **Output:** NumPy classifier with manual backprop

Day 31 - Keras - First Real Neural Network

- **Topic taught:** Keras anatomy - Sequential vs Functional API; compile, fit, evaluate; loading MNIST
- **In class:** Live build - MNIST classifier with a dense network in Keras; hit 97%+
- **Homework:** Push MNIST accuracy above 98%; document what changes worked
- **Output:** 98%+ MNIST classifier on GitHub

Day 32 - Overfitting, Dropout, Batch Norm, Early Stopping

- **Topic taught:** Reading training vs validation loss curves; dropout, batch normalization, early stopping callbacks
- **In class:** Deliberately overfit a model, then fix it with dropout; see the curves change live
- **Homework:** Train 4 versions of MNIST - vanilla, with dropout, with batch norm, with both; plot all 4 loss curves on one chart
- **Output:** 4-version comparison notebook

Day 33 - CNNs - Convolution & Pooling

- **Topic taught:** Convolution as 'a filter that slides over an image and detects patterns'; MaxPooling; computing output dimensions
- **In class:** Build first CNN on CIFAR-10; achieve 60%+
- **Homework:** Push CIFAR-10 accuracy higher by adding more conv blocks; aim for 70%+
- **Output:** Custom CNN on CIFAR-10

Day 34 - CNN Architectures & Transfer Learning

- **Topic taught:** History of LeNet, VGG, ResNet, MobileNet; what transfer learning is; freezing layers, fine-tuning the head
- **In class:** Live transfer learning with MobileNetV2 on a cats-vs-dogs subset
- **Homework:** Apply transfer learning to a Kaggle image classification dataset; aim for 90%+ accuracy
- **Output:** Transfer-learned classifier on Kaggle dataset

Day 35 - Image Augmentation & Data Pipelines

- **Topic taught:** Augmentation is 'free data'; rotation, flip, zoom, brightness; ImageDataGenerator and tf.data
- **In class:** Visualize augmentation effects; build an augmentation pipeline
- **Homework:** Compare model accuracy with vs without augmentation on Day 34 project; document the difference
- **Output:** Augmented training pipeline + comparison

Day 36 - RNNs & LSTMs (Sequence Modeling)

- **Topic taught:** Why sequence data is different from images; vanishing gradients; LSTM as the solution
- **In class:** Train an LSTM on the IMDB sentiment dataset
- **Homework:** Train a sentiment classifier on a tweets dataset; compare to a TF-IDF + logistic regression baseline
- **Output:** Sentiment classifier with LSTM vs baseline

Day 37 - Time Series Forecasting with Deep Learning

- **Topic taught:** How to frame time-series data for supervised learning; LSTM for forecasting
- **In class:** Forecast stock prices with LSTM; compare to a moving-average baseline
- **Homework:** Forecast one real-world time series of your choice; report MAE and plot predictions
- **Output:** Time series forecasting notebook

Day 38 - PROJECT 3 - Day 1: Plan & Baseline

- **Topic taught:** Project briefing; students choose a custom image classification problem (dog breeds, plant diseases, food categories, etc.)
- **In class:** Collect/curate the dataset; train a baseline model
- **Homework:** Get baseline accuracy logged; document validation accuracy
- **Output:** Baseline image classifier trained

Day 39 - PROJECT 3 - Day 2: Build & Tune

- **Topic taught:** Apply transfer learning + augmentation + tuning

- **In class:** Build session - improve baseline with transfer learning
- **Homework:** Wrap model into a Gradio app running locally
- **Output:** Gradio app working locally

Day 40 - PROJECT 3 - Day 3: Deploy & Present

- **Topic taught:** Deploying to Hugging Face Spaces (free, with GPU)
- **In class:** Live HF Spaces deployment + 5-minute presentations
- **Homework:** Polish README; pin project on GitHub
- **Output:** PORTFOLIO PROJECT 3 - live image classifier on HF Spaces

Phase 4 - Computer Vision (Day 41 to 46)

Day 41 - OpenCV Essentials

- **Topic taught:** Reading and writing images and video; filters (blur, sharpen, Canny edge detection); color space conversion, resizing, cropping, rotation
- **In class:** Live demo of 5 filters and transformations
- **Homework:** Build a CLI Photo Editor with 5 operations; push with sample inputs and outputs
- **Output:** Photo editor on GitHub

Day 42 - YOLO Object Detection

- **Topic taught:** Detection vs classification; YOLOv8 from Ultralytics; filtering by class and confidence
- **In class:** Live YOLOv8 demo on images and webcam feed
- **Homework:** Build a 'vehicle counter from traffic video' script
- **Output:** Vehicle counter script + demo video

Day 43 - Image Segmentation & MediaPipe Pose Tracking

- **Topic taught:** Semantic vs instance segmentation; SAM (Segment Anything); MediaPipe for pose, hand, face tracking
- **In class:** Run SAM live; MediaPipe pose detection on webcam
- **Homework:** Build a real-time webcam app that does ONE useful thing (pushup counter, gesture controller, attention tracker, etc.)
- **Output:** Real-time webcam app

Day 44 - PROJECT 4 - Day 1: Plan

- **Topic taught:** Project briefing; idea bank - face attendance, document OCR, retail shelf analyzer, traffic counter, etc.
- **In class:** Choose problem; start prototype
- **Homework:** Working prototype tonight
- **Output:** CV prototype

Day 45 - PROJECT 4 - Day 2: Build

- **Topic taught:** Iterate; wrap in Streamlit or Gradio
- **In class:** Build session
- **Homework:** Demo-ready app + record a 30-second screen recording
- **Output:** Demo-ready CV app

Day 46 - PROJECT 4 - Day 3: Deploy & Present

- **Topic taught:** Deploy to HF Spaces or Streamlit Cloud
- **In class:** Deployment + presentations
- **Homework:** Update LinkedIn featured section with demo link
- **Output:** PORTFOLIO PROJECT 4 - live CV application

Phase 5 - NLP & GenAI / LLMs (Day 47 to 54)

Day 47 - Text Preprocessing & TF-IDF

- **Topic taught:** Tokenization, lowercasing, stopwords removal, lemmatization; TF-IDF vectorization; baseline text classification
- **In class:** Live build - spam classifier with TF-IDF + Logistic Regression
- **Homework:** Achieve 95%+ on the SMS Spam dataset using only TF-IDF and Logistic Regression
- **Output:** Spam classifier notebook

Day 48 - Word Embeddings (Word2Vec, GloVe)

- **Topic taught:** Embeddings intuition; the 'king - man + woman = queen' example; loading pre-trained GloVe
- **In class:** Explore embedding space - find similar words, do analogies
- **Homework:** Build a 'find similar words' tool using GloVe
- **Output:** Word similarity tool

Day 49 - Transformers & Attention (Intuition Only)

- **Topic taught:** Attention as 'looking at the right words at the right time'; transformer block at high level; encoder vs decoder vs encoder-decoder; BERT vs GPT vs T5 in one slide each
- **In class:** Load BERT from Hugging Face and inspect outputs
- **Homework:** Read 'The Illustrated Transformer' blog; write 5 questions you still have
- **Output:** BERT inspection notebook + reflection questions

Day 50 - Hugging Face Pipeline API

- **Topic taught:** The pipeline API for classification, NER, summarization, translation; loading any model from the Hub
- **In class:** Demo of all 4 pipelines on custom text
- **Homework:** Build a Multi-Tool Text Analyzer in Streamlit - takes any text, runs 4 NLP tasks on it, deploys live
- **Output:** Live multi-tool text analyzer

Day 51 - LLM Landscape

- **Topic taught:** GPT, Claude, Gemini, Llama, Mistral, Qwen - sizes, hosting, costs; open weights vs closed API; free options (Groq, HF Inference)
- **In class:** Set up API access; make first LLM API call
- **Homework:** Make 10 API calls across 3 different LLMs with the same prompt; compare responses
- **Output:** LLM comparison notebook

Day 52 - Prompt Engineering

- **Topic taught:** Zero-shot, one-shot, few-shot prompting; chain-of-thought; structured output prompting; system messages and role assignment
- **In class:** Live before/after - bad prompt vs engineered prompt on the same task
- **Homework:** Build a Prompt Library of 10 reusable prompts for common business tasks (extraction, classification, summarization, etc.)
- **Output:** Prompt library repo

Day 53 - RAG - Chunking, Embeddings, Vector Stores

- **Topic taught:** Why RAG exists; full pipeline - chunk, embed, store, retrieve, augment, generate; vector stores (Chroma, FAISS)
- **In class:** Live RAG build - load a PDF, chunk it, embed, store, query
- **Homework:** Build a 'Chat With My Resume' RAG app
- **Output:** Personal resume chatbot

Day 54 - PROJECT 5 - Domain-Specific RAG Chatbot

- **Topic taught:** Building a real RAG application end-to-end
- **In class:** Choose domain (legal docs, medical FAQs, internal wiki, study notes, customer support); build the full pipeline; add Streamlit chat UI with history; deploy
- **Homework:** Polish README; record demo video
- **Output:** PORTFOLIO PROJECT 5 - live deployed RAG chatbot

Phase 6 - MLOps & Deployment (Day 55 to 58)

Day 55 - FastAPI - Wrapping Models as Services

- **Topic taught:** Why FastAPI beats Flask for ML; Pydantic for input validation; auto-generated /docs page
- **In class:** Wrap Day 28 model in a FastAPI endpoint; test with curl and Postman
- **Homework:** Get the API running locally with 3 documented endpoints
- **Output:** Model behind a REST API

Day 56 - Docker for ML

- **Topic taught:** Docker fundamentals - image, container, registry; reading and writing a Dockerfile
- **In class:** Dockerize yesterday's FastAPI service; build, run, push to Docker Hub
- **Homework:** Public Docker image on Docker Hub; add docker run instructions to README
- **Output:** Containerized service

Day 57 - Cloud Deployment & MLflow Tracking

- **Topic taught:** Free cloud tiers (Render, Railway, HF Spaces); MLflow for experiment tracking
- **In class:** Deploy Docker container live to cloud; set up MLflow tracking with 5 runs
- **Homework:** Live cloud URL of your service in README + MLflow dashboard screenshot
- **Output:** Live cloud-deployed service

Day 58 - Model Monitoring & CI/CD with GitHub Actions

- **Topic taught:** Model drift, monitoring concepts; a simple GitHub Actions workflow that runs tests on push
- **In class:** Walk through a sample workflow file; add CI to one repo
- **Homework:** Green CI check on at least one repository
- **Output:** CI workflow on GitHub

Phase 7 - Capstone & Career (Day 59 to 60)

Day 59 - PROJECT 6 (Capstone) Presentations

- **Topic taught:** Presenting technical projects to non-technical audiences
- **In class:** Each student presents capstone in 8-10 minutes - problem, data, method, results, live demo, next steps
- **Homework:** Apply all feedback overnight; polish capstone repo
- **Output:** PORTFOLIO PROJECT 6 - polished capstone

Day 60 - Resume, LinkedIn, Mock Interviews, Road Ahead

- **Topic taught:** Resume writing for AI/ML roles; LinkedIn profile optimization (headline, about, featured, experience, skills); common interview questions
- **In class:** Resume + LinkedIn walkthrough; mock interviews in pairs (rotation); 90-day post-program roadmap
- **Homework:** Apply to 5 jobs this week; continue with Kaggle competitions
- **Output:** Job-application-ready profile + 90-day plan

4. The 6 Portfolio Projects

Every student finishes the program with 6 deployed projects on their public GitHub. These are what get them interviews.

Project 1 - Real-World Exploratory Data Analysis

Day 12

What it is	Student picks a real dataset (Airbnb listings, IMDb movies, NYC taxi, etc.), cleans the data, builds 8-10 visualizations, and writes a clear narrative of what the data reveals. The classic first task every data analyst does on the job.
Skills used	Python, Pandas, Matplotlib, Seaborn, data storytelling, GitHub

Project 2 - End-to-End Classical ML App

Day 26-28

What it is	Student picks a business problem (customer churn, loan default, house prices, or employee attrition), builds and tunes 3+ ML models, selects the best one, wraps it in a Streamlit web app where anyone can enter inputs and get predictions, and deploys it live on Streamlit Cloud.
Skills used	scikit-learn, XGBoost, feature engineering, model evaluation, Streamlit, deployment

Project 3 - Image Classifier with Transfer Learning

Day 38-40

What it is	Student picks a custom image classification problem (dog breeds, plant diseases, food categories, etc.), curates a small dataset, applies transfer learning with a pre-trained CNN, builds a Gradio app where users upload an image and get a prediction, and deploys it on Hugging Face Spaces.
Skills used	Keras / TensorFlow, CNNs, transfer learning, Gradio, Hugging Face Spaces

Project 4 - Computer Vision Application

Day 44-46

What it is	Student builds a deployed CV app using pre-trained models. Examples: vehicle counter from CCTV footage, attendance system via face recognition, document OCR, pose-tracking fitness app. Wrapped in Streamlit or Gradio and deployed publicly.
Skills used	OpenCV, YOLO, MediaPipe, Streamlit / Gradio, deployment

Project 5 - Domain-Specific RAG Chatbot

Day 54

What it is	Student picks a domain (legal contracts, medical FAQs, company docs, study notes), uploads PDFs, builds a full Retrieval-Augmented Generation pipeline: chunk + embed + store in vector DB + retrieve + answer with citations. Streamlit chat UI, deployed live. The single most in-demand AI skill in 2026.
Skills used	LLM APIs, embeddings, vector databases (Chroma / FAISS), LangChain, Streamlit

Project 6 - Capstone Project (student's choice)

Day 55-60

What it is	The final showcase. Student picks one capstone idea, works on it throughout Phase 6, and presents on Day 59. Requirements: real dataset, trained model, deployed UI with public link, clean README, demo video. This is the project they lead with in interviews.
Skills used	Everything combined - chosen and applied independently by the student

5. Capstone Project Options

Students choose one of these 10 ideas. Each is calibrated to be impressive in an interview without being unrealistic for 60 days.

Project	Description
Resume Screening Assistant	Ranks candidate resumes against a job description using RAG + classification; explains why each was ranked
Loan Default Predictor with Explainability	End-to-end ML app with SHAP explanations for each prediction
Medical X-Ray Triage	CNN for pneumonia detection from chest X-rays with confidence score
Regional Language Sentiment Analyzer	Sentiment classifier for Hindi / Tamil / Telugu social media using multilingual transformers
Face Recognition Attendance System	OpenCV + face recognition + Streamlit UI; logs attendance to CSV
E-Commerce Recommendation Engine	Collaborative filtering + content-based hybrid with LLM-generated explanations
Stock Movement Forecaster	LSTM model + classical baseline with honest reporting of model limitations
Chat With Company Docs (RAG)	Upload PDFs, ask questions, get cited answers - generalizable to many real jobs
Real-Time Traffic Counter	YOLOv8 + object tracking; counts vehicles by type with Streamlit dashboard
Personal Finance Categorizer	Upload bank statement CSV, auto-categorize transactions, show spending dashboard

6. Tools & Tech Stack

All tools used are free and industry-standard.

Category	Tools used
Language	Python 3.10+
Environment	Anaconda, VS Code, Jupyter, Google Colab (free GPU)
Data libraries	NumPy, Pandas, Matplotlib, Seaborn
Classical ML	scikit-learn, XGBoost, LightGBM, imbalanced-learn
Deep Learning	TensorFlow / Keras
Computer Vision	OpenCV, Ultralytics YOLOv8, MediaPipe
NLP & LLMs	Hugging Face Transformers, Sentence-Transformers, LangChain, ChromaDB / FAISS
Deployment	FastAPI, Streamlit, Gradio, Docker, MLflow, GitHub Actions
Hosting (free tiers)	GitHub, Streamlit Cloud, Hugging Face Spaces, Render, Docker Hub

7. What Every Student Will Have on Day 60

A complete public GitHub profile with 6 pinned projects
4 live deployed web apps (Streamlit Cloud / Hugging Face Spaces / Render)
1 capstone project with public demo and 3-minute walkthrough video
A polished 1-page resume tailored to junior AI / ML / Data Analyst roles
A complete LinkedIn profile with featured projects and demo links
Proven ability to take a dataset and build, evaluate, and deploy a model end-to-end
Mock-interview clearance: can explain every project in plain language and answer common ML interview questions